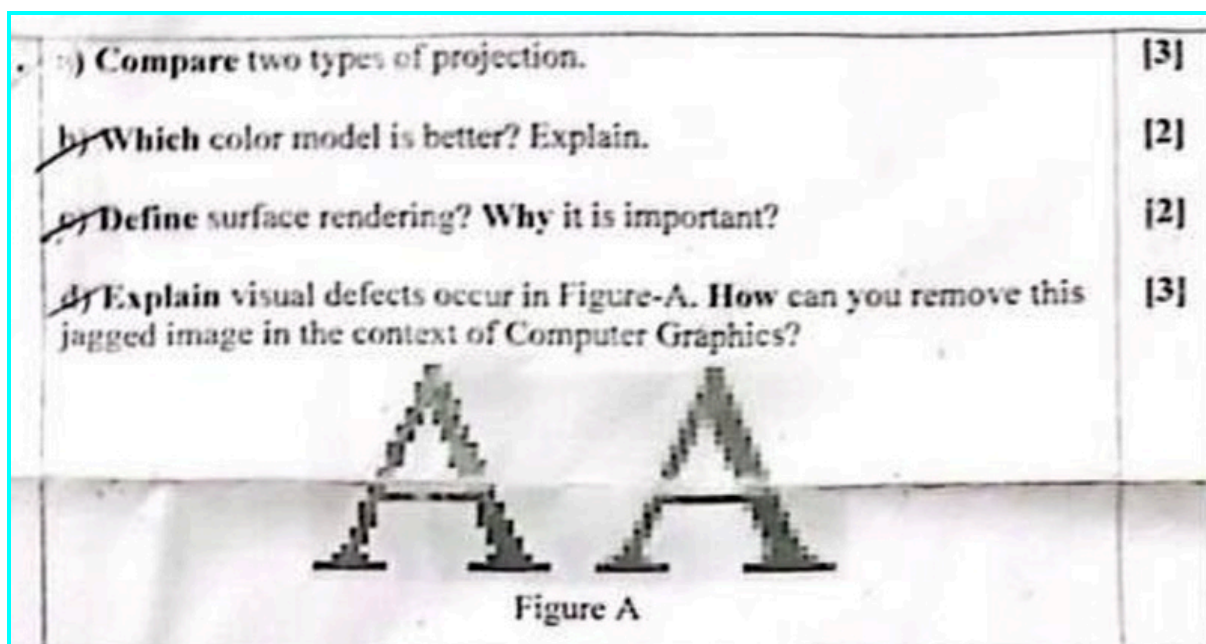




Question: Compare two types of projection. [Marks: 3]

(উত্তর: Two main types of projection are Parallel Projection and Perspective Projection. Below is a comparison between them:)

Parameter	Parallel Projection	Perspective Projection
Center of Projection	প্রজেকশন সেন্টার অসীম দূরত্বে থাকে (Located at infinity)।	প্রজেকশন সেন্টার নির্দিষ্ট সসীম দূরত্বে থাকে (Located at a finite distance)।
Object Size	বস্তুর আসল সাইজ ও শেপ বজায় থাকে (Preserves true relative dimensions)।	বস্তুর দূরত্ব বাড়লে তা ছোট দেখায় (Distance causes objects to appear smaller)।
Realism	কম রিয়েলিস্টিক, ইঞ্জিনিয়ারিং ড্রইংয়ে বেশি ব্যবহৃত হয় (Less realistic, used in technical drawings)।	অত্যন্ত রিয়েলিস্টিক, মানব চোখের দেখার মতো (Highly realistic, resembles human vision)।

Question: Which color model is better? Explain. [Marks: 2]

(উত্তর: No single color model is universally "better"; choice depends on application context—RGB for displays, CMYK for printing, and HSV/HSI for human perception and computer vision.)

- ডিজিটাল ডিসপ্লেসের জন্য (**RGB Model**): মনিটর, মোবাইল বা টিভি স্ক্রিনে আলো তৈরি করে কালার দেখানোর জন্য এটি সেরা (Additive color model used for active light sources)।
- প্রিন্টিং বা ছাপাখানার জন্য (**CMYK Model**): কাগজে কালি বা রঞ্জক পদার্থ দিয়ে প্রিন্ট করার জন্য এটি ব্যবহৃত হয় (Subtractive color model used for physical printing)।
- হিউম্যান পারসেপশন বা ইমেজ প্রসেসিংয়ের জন্য (**HSV/HSI Model**): মানুষের চোখ যেভাবে রঙ, উজ্জ্বলতা অনুধাবন করে তার সাথে সামঞ্জস্যপূর্ণ (Aligns with human perception of hue, saturation, and brightness)।

Question: Define surface rendering? Why it is important? [Marks: 2]

(উত্তর: **Surface rendering is the process of applying shading, texture, color, and lighting models to a 3D geometric object's surface to create a realistic 2D image.**)

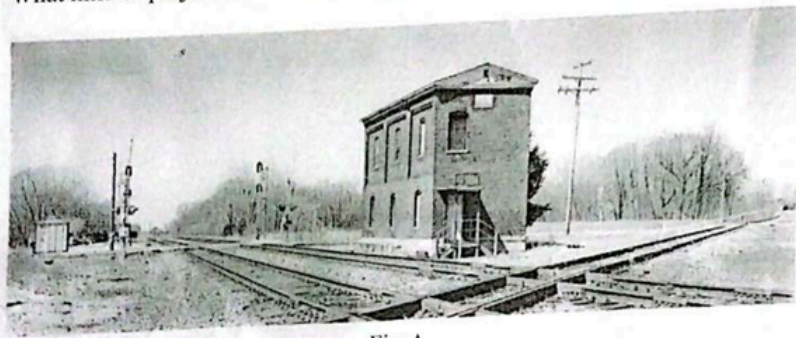
- সংজ্ঞা (**Definition**): একটি ৩ডি অবজেক্টের উপরিভাগে লাইটিং, শেডিং, শেড বা টেক্সচার প্রয়োগ করে তাকে দ্বিমাত্রিক (2D) স্ক্রিনে ফুটিয়ে তোলার প্রক্রিয়াকে Surface Rendering বলে।
- গুরুত্ব (**Importance**): এটি ছাড়া ৩ডি অবজেক্টকে সমতল ওয়ারফ্রেমের (wireframe) মতো মনে হয়। Surface rendering অবজেক্টে গভীরতা (depth perception), রিয়েলিজম এবং লাইটের প্রতিফলন ফুটিয়ে তুলতে সাহায্য করে।

Question: Explain visual defects occur in Figure-A. How can you remove this jagged image in the context of Computer Graphics? [Marks: 3]

(Scenario: Figure-A shows two instances of the letter 'A'—the left one exhibits pixelated stair-step artifacts, and the right one is smoothed.)

(উত্তর: **The visual defect in Figure-A (left letter 'A') is called Aliasing or "Jaggies". It can be removed using Anti-aliasing techniques.**)

- ভিজ্যুয়াল ডিফেক্টের কারণ (**Explanation of Visual Defect**):
- চিত্রের বাম পাশের 'A' অক্ষরে সিঁড়ির মতো ধারালো বাঁক বা পিক্সেলের ধাপ দেখা যাচ্ছে। কম রেজোলিউশনের রাস্টার গ্রিডে বক্ররেখা বা কোনাকুনি লাইনকে স্যাম্পলিং করার সময় পিক্সেলগুলো পুরোপুরি অন বা অফ হওয়ার কারণে এই **Aliasing** বা **Jagged edges** তৈরি হয়।
- সমস্যা সমাধানের উপায় (**How to Remove**):
- এই খসখসে বা সিঁড়ির মতো ভাব দূর করতে **Anti-aliasing** টেকনিক ব্যবহার করা হয় (যেমনটি ডান পাশের 'A'-তে দেখা যাচ্ছে)। Anti-aliasing পদ্ধতিতে বাউন্ডারি পিক্সেলগুলোতে ইনটেনসিটি ফিল্টারিং বা গ্রেডিং (Grey-scale pixel shading) করা হয়, যা লাইনগুলোকে মসৃণ ও স্পষ্ট দেখায়।

1.	(a)	What kind of projection is shown in figure A? Describe each of with examples.	[4+4]	CO1
				
		Fig:A		
(b)	A small e-learning company wants to create a 20-second animated clip explaining how a plant grows. The designer has completed the rough sketches but is unsure how to proceed with converting them into smooth animation. Explain which steps of the animation sequence design process the designer should follow next and how those steps will help transform the sketches into a smooth final animation.			[8]

Question 1(a) — [4 Marks]

What kind of projection is shown in Figure A? Describe each of with examples.

Answer:

The projection shown in Figure A is **Perspective Projection**.

In perspective projection, the projection lines converge at a **single point called the center of projection**. Objects that are farther from the viewer appear smaller, while objects closer to the viewer appear larger.

Example:

A railway track appears to become narrower as it goes farther away from the observer, even though the two rails are actually parallel.

(বাংলায়: **Perspective projection**-এ দূরের জিনিস ছোট দেখায় এবং কাছের জিনিস বড় দেখায়। **Railway track**-এর দুইটা **parallel rail** দূরে গিয়ে এক জায়গায় মিশে যাচ্ছে মনে হয়—এটাই **perspective effect**.)

Main characteristics:

1. Projection lines converge at one point.
2. Distant objects appear smaller.
3. Parallel lines may appear to meet at a vanishing point.

(বাংলায়: সব **projection line** একটা **point**-এর দিকে যায়, দূরের **object** ছোট হয় এবং **parallel line** দূরে গিয়ে মিলেছে মনে হতে পারে।)

Question 1(b) — [4 Marks]

A small e-learning company wants to create a 20-second animated clip explaining how a plant grows. The designer has completed the rough sketches but is unsure how to proceed with converting them into smooth animation. Explain which steps the designer should follow next and how those steps will help transform the sketches into a smooth final animation.

Answer:

After completing the rough sketches, the designer should follow these steps:

1. Clean-up and Final Drawing:

The rough sketches should be refined and converted into clean, consistent drawings.

(বাংলায়: **Rough drawing**-গুলোকে পরিষ্কার ও সুন্দর **final drawing**-এ **convert** করতে হবে।)

2. In-betweening:

The designer should create intermediate frames between the keyframes.

(বাংলায়: দুইটা **important frame**-এর মাঝখানে আরও **frame** বানাতে হবে, যাতে **plant**-এর **growth** হঠাৎ না হয়ে **smoothly** হয়।)

3. Timing:

The frames should be arranged with proper timing according to the 20-second duration.

(বাংলায়: কোন **movement** কত দ্রুত হবে সেটা ঠিক করতে হবে।)

4. Preview and Refinement:

The animation should be previewed and necessary adjustments should be made to movement, timing, and transitions.

(বাংলায়: **Animation** চালিয়ে দেখে যেখানে **movement jerky** বা **unnatural** লাগে সেখানে **correction** করতে হবে।)

Therefore, these steps convert the rough sketches into a smooth and visually consistent final animation.

(বাংলায়: অর্থাৎ **clean-up** → **in-betweening** → **timing** → **preview/refinement** করলে **rough sketch** থেকে **smooth final animation** পাওয়া যাবে।)

outcomes. All portions of each question must be answered sequentially.)

1.	a)	Suppose Ms. Ahmed is planning to start a digital marketing agency where she will create digital content such as online ads, social media graphics, and website designs. She is unsure about which color model to employ for her projects. Could you advise her on the best color model to use for her digital content creation and explain why?	[5]	CO1
	b)	Suppose an architect, Ms. Thompson, is utilizing 3D modeling software to create visualizations of her building designs for client presentations. She needs to choose between perspective and parallel projection techniques to best represent the spatial relationships and dimensions in her models. Now explain the differences between perspective and parallel projection, highlighting their respective advantages and disadvantages for architectural visualization?	[3]	

Question 1(a) — [5 Marks]

Full Question with Scenario:

Suppose Ms. Ahmed is planning to start a digital marketing agency where she will create digital content such as online ads, social media graphics, and website designs. She is unsure about which color model to employ for her projects. Could you advise her on the best color model to use for her digital content creation and explain why?

Answer:

The best color model for Ms. Ahmed's digital content creation is the **RGB (Red, Green, Blue) color model**.

RGB is an **additive color model** that uses three primary colors: **Red, Green, and Blue**. Different intensities of these three colors are combined to produce a wide range of colors.

Why RGB is suitable:

1. Designed for digital screens:

RGB is mainly used for electronic displays such as computer monitors, smartphones, tablets, and televisions.

(বাংলায়: **Digital content** যেহেতু **screen-এ** দেখা হবে, তাই **RGB** সবচেয়ে **suitable**.)

2. Suitable for web and social media:

Websites, online advertisements, social media graphics, and other digital content are displayed using RGB colors.

(বাংলায়: **Facebook/Instagram graphics, website design, online ads**—সব **screen-based** হওয়ায় **RGB** ব্যবহার করা হয়।)

3. Wide range of colors:

By changing the intensity of Red, Green, and Blue, many different colors can be produced.

(বাংলায়: **Red, Green** আর **Blue**-এর **intensity** পরিবর্তন করে অনেক ধরনের **color** তৈরি করা যায়।)

4. **Additive color mixing:**

When all three colors are combined at maximum intensity, they produce **white**; when their intensity is zero, they produce **black**.

(বাংলায়: **R + G + B full intensity = White**, আর তিনটার **intensity 0 = Black**.)

Example:

A website banner or Instagram advertisement designed for viewing on a smartphone should be created using the **RGB color model**.

Therefore, **RGB is the most appropriate color model for Ms. Ahmed's digital content creation because her work is primarily intended for digital screens.**

Question 1(b) — [3 Marks]

Full Question with Scenario:

Suppose an architect, Ms. Thompson, is utilizing 3D modeling software to create visualizations of her building designs for client presentations. She needs to choose between perspective and parallel projection techniques to best represent the spatial relationships and dimensions in her models. Now explain the differences between perspective and parallel projection, highlighting their respective advantages and disadvantages for architectural visualization.

Answer:

Perspective Projection and **Parallel Projection** are two common projection techniques used in computer graphics.

Perspective Projection	Parallel Projection
Projection lines converge at a point called the center of projection .	Projection lines remain parallel to each other.
Distant objects appear smaller.	Object size does not change with distance.
Gives a more realistic view.	Gives a more accurate representation of dimensions.

Useful for realistic architectural presentations.

Useful for technical drawings and measurements.

(বাংলায়: **Perspective**-এ দূরের **object** ছোট দেখায়, তাই **realistic** লাগে। **Parallel**-এ দূরের **object** ছোট হয় না, তাই **actual size/dimension** বোঝানোর জন্য ভালো।)

Perspective Projection

Advantages:

- Provides a realistic and natural-looking view.
- Shows depth effectively.

(বাংলায়: মানুষের চোখে যেমন বাস্তবে দেখা যায়, অনেকটা তেমন দেখায়।)

Disadvantages:

- Measurements and dimensions may appear distorted.
- Distant objects appear smaller.

(বাংলায়: দূরের জিনিস ছোট হয়ে যাওয়ায় **exact measurement** বোঝানো কঠিন।)

Parallel Projection

Advantages:

- Preserves relative dimensions and proportions.
- Suitable for architectural and engineering drawings.

(বাংলায়: **Building**-এর **length, width, height** তুলনামূলকভাবে **accurately** দেখানো যায়।)

Disadvantages:

- Less realistic than perspective projection.
- Depth may be harder to perceive.

(বাংলায়: দেখতে **realistic** কম লাগে এবং কোন **object** কত দূরে সেটা বোঝা কঠিন হতে পারে।)

Conclusion:

For **realistic client presentations**, Ms. Thompson should prefer **Perspective Projection**, while **Parallel Projection** is better when accurate dimensions and technical details are more important.

(বাংলায়: **Client**-কে **building** বাস্তবে কেমন দেখাবে দেখাতে চাইলে **Perspective** ভালো; আর **technical drawing/dimension** দেখাতে চাইলে **Parallel** ভালো।)

Outcomes: All portions of each question must be answered sequentially.

1.	a)	State the differences between additive and subtractive color models and one practical use of the subtractive color model.	[3]	CO1
	b)	An animation studio is facing jagged edges in its 3D-rendered scenes, making the visuals look less realistic. They want to smooth edges but are unsure how to do so. Explain different techniques to reduce jagged edges, including their benefits and drawbacks.	[5]	

Question 1(a) — [3 Marks]

Full Question:

State the differences between additive and subtractive color models and one practical use of the subtractive color model.

Answer:

The main differences between **additive** and **subtractive** color models are:

Additive Color Model	Subtractive Color Model
Uses Red, Green, and Blue (RGB) as primary colors.	Uses Cyan, Magenta, and Yellow (CMY) as primary colors.
Colors are produced by adding light .	Colors are produced by subtracting light .
Combining all primary colors produces white .	Combining all primary colors ideally produces black/dark .
Mainly used for digital displays .	Mainly used for printing .

(বাংলায়: **Additive model**-এ আলো যোগ করে **color** তৈরি হয়—**RGB**। **Subtractive model**-এ **ink/pigment** দিয়ে আলো শোষণ বা **subtract** করে **color** তৈরি হয়—**CMY**।)

Practical Use:

A practical use of the **subtractive color model** is **color printing**, where cyan, magenta, yellow and usually black (**CMYK**) inks are used.

(বাংলায়: **Printer**-এ যে **ink** ব্যবহার করে **paper**-এ **color print** করা হয়, সেখানে **subtractive color model** ব্যবহার হয়।)

Question 1(b) — [5 Marks]

Full Question with Scenario:

An animation studio is facing jagged edges in its 3D-rendered scenes, making the visuals look less realistic. They want to smooth edges but are unsure how to do so. Explain different techniques to reduce jagged edges, including their benefits and drawbacks.

Answer:

The jagged or stair-step appearance of edges in computer graphics is called **aliasing**. The process of reducing these jagged edges is called **anti-aliasing**.

Several techniques can be used to reduce jagged edges:

1. Supersampling Anti-Aliasing (SSAA)

The scene is rendered at a higher resolution and then downsampled to the required resolution.

Benefit: Produces very smooth and high-quality edges.

Drawback: Requires more processing time and memory, so it is computationally expensive.

(বাংলায়: আগে বড় **resolution**-এ **image** বানিয়ে পরে ছোট করা হয়। **Quality** অনেক ভালো হয়, কিন্তু **processing** বেশি লাগে।)

2. Multisample Anti-Aliasing (MSAA)

MSAA samples multiple points around the edges of polygons and uses these samples to produce smoother boundaries.

Benefit: Provides good edge smoothing with less computational cost than SSAA.

Drawback: Mainly works on polygon edges and may not smooth all types of image details.

(বাংলায়: **Edge**-এর আশেপাশে **multiple sample** নিয়ে **smooth** করে। **SSAA**-এর চেয়ে কম **costly**, কিন্তু সব ধরনের **detail smooth** করতে পারে না।)

3. Fast Approximate Anti-Aliasing (FXAA)

FXAA is a post-processing technique that detects jagged edges in the rendered image and smooths them.

Benefit: Very fast and requires relatively low processing power.

Drawback: It may make the image slightly blurry and can reduce fine details.

(বাংলায়: **Rendering** শেষ হওয়ার পরে **jagged edge detect** করে **smooth** করে। খুব **fast**, কিন্তু **image** একটু **blurry** হতে পারে।)

4. Temporal Anti-Aliasing (TAA)

TAA uses information from multiple frames over time to reduce jagged edges, especially in animations.

Benefit: Works well for moving scenes and can provide smooth visual results.

Drawback: May produce ghosting or blurry effects when objects move quickly.

(বাংলায়: আগের ও বর্তমান **frame**-এর **information** ব্যবহার করে **smooth** করে। **Animation**-এর জন্য ভালো, কিন্তু **fast movement** হলে **ghosting** হতে পারে।)

Conclusion:

For a **3D animation studio**, techniques such as **MSAA**, **FXAA**, and **TAA** can be used depending on the required quality and rendering speed. **SSAA** provides the highest quality but requires more computational resources.

(বাংলায়: **Best quality** চাইলে **SSAA**, **performance**-এর সাথে **quality balance** চাইলে **MSAA**, খুব **fast solution** চাইলে **FXAA**, আর **animation**-এর জন্য **TAA** useful।)

outcomes. All portions of each question must be answered sequentially.]

1.	A student is assigned to create a short animation of a car moving on a road for a magazine advertisement. Before starting, he carefully plans the animation sequence step by step. Since the final output will be printed in a magazine that only supports black ink, the designer also applies a technique using small dots of varying sizes to represent different shades in the images.	[9]	CO1
	a) List and Explain the steps for designing an animation sequence?		
	b) Identify which technique is being used for printing?		
	c) Explain how do varying dot sizes represent intensity levels?		

Question 1 — Exam Style Answer

Full Question with Scenario

[9 Marks]

A student is assigned to create a short animation of a car moving on a road for a magazine advertisement. Before starting, he carefully plans the animation sequence step by step. Since the final output will be printed in a magazine that only supports black ink, the designer also applies a technique using small dots of varying sizes to represent different shades in the images.

- a) List and explain the steps for designing an animation sequence.
 - b) Identify which technique is being used for printing.
 - c) Explain how varying dot sizes represent intensity levels.
-

(a) List and Explain the Steps for Designing an Animation Sequence

1. Storyboard – Plan the complete animation visually by showing the important scenes and actions in sequence.

(Storyboard = পুরো animation-এর scene-by-scene পরিকল্পনা।)

2. Key Frames – Define the important starting, ending, and major positions of the objects.

(Key frame = গুরুত্বপূর্ণ অবস্থান/মুহূর্ত নির্ধারণ করা।)

3. In-between Frames – Create intermediate frames between the key frames to produce smooth movement.

(In-between = দুইটি key frame-এর মাঝের movement তৈরি করা।)

4. Timing and Sequence – Determine the order and duration of each frame so that the animation moves at the desired speed.

(Timing = কোন frame কতক্ষণ থাকবে এবং movement কত দ্রুত হবে তা নির্ধারণ করা।)

5. Rendering / Final Output – Generate the final sequence by displaying the frames continuously to produce the animation.

(Rendering = final animation output তৈরি করা।)

(b) Identify the Technique Used for Printing

The technique used is **Halftoning**.

(**Halftoning** = ছোট ছোট **black dots**-এর **size/arrangement** পরিবর্তন করে বিভিন্ন **shade** তৈরি করার **printing technique**।)

(c) How Do Varying Dot Sizes Represent Intensity Levels?

In halftoning, different intensity levels are represented by changing the **size or density of black dots**.

- **Large dots** cover more area with black ink, producing a **darker or higher-intensity** region.
(বড় **dot** = বেশি **black area** = বেশি **dark**।)
- **Small dots** cover less area with black ink, producing a **lighter or lower-intensity** region.
(ছোট **dot** = কম **black area** = বেশি **light**।)
- **Intermediate dot sizes** produce **intermediate gray shades**.
(মাঝারি **dot** = মাঝারি **gray shade**।)

Thus, although only black ink is used, varying the dot size creates the visual appearance of different **gray intensity levels**.

★ Very Short Revision

(a) Storyboard → Key Frames → In-between Frames → Timing/Sequence → Rendering

(b) **Halftoning**

(c) Large dots → Darker, Small dots → Lighter, Medium dots → Intermediate gray.